AI-Driven Delinquency Risk Insights and Recommendation Report

> **Project:** AI-Driven Delinquency Risk Assessment for Geldium's Credit Card Collections
> **Document 3 of 4:** Insights & Recommendations
> **Audience:** Head of Collections, Geldium

Purpose

This report turns the findings of the EDA and the predictive model into **business action**: which customers are most at risk, what drives that risk, what the Collections team should do about it, and how to do it responsibly.

1. Summary of Predictive Insights

1.1 High-Risk Customer Segments

Analysis of Geldium's dataset shows three segments with a heightened likelihood of missing credit card payments:

Segment	Profile	Why It's Risky
Over-utilisers	Credit utilisation **above 90%**, especially with low income or unstable employment	Financially overextended with little buffer
Repeat missers	**Multiple missed payments in the past 3 months**	Shows a recurring delinquency pattern
Young, thin-file borrowers	**Under age 30**, limited credit history, high debt-to-income ratio	Little track record and high debt load

These groups are the ideal targets for **proactive interventions and outreach**.

1.2 Top 3 Risk Factors

1. **High Credit Utilisation Ratio (>90%)**
   Indicates financial overextension and correlates strongly with delinquency.
2. **Recent Missed Payments**
   Payment history over the past 3 months is the **most reliable short-term predictor** of future risk.
3. **High Debt-to-Income Ratio (>0.6)**
   Signals limited repayment capacity, especially when paired with rising balances.

2. Recommendation Framework

2.1 SMART Recommendation

> **Goal:** Launch a **6-week SMS and email outreach campaign** targeting **high-risk customers under 30 with 2+ missed payments in the past 3 months**, aiming to **reduce 30-day delinquency in this group by 10%**.

SMART Element	How the Recommendation Meets It
Specific	Targets one clearly defined segment: under-30s with 2+ missed payments in 3 months
Measurable	10% reduction in 30-day delinquency within the segment
Achievable	Uses low-cost, existing channels (SMS and email)
Relevant	Focuses on the segment where early intervention has the highest impact
Time-bound	Runs for 6 weeks

2.2 Rationale

- **Specific and actionable** — focused on a segment where early intervention has high impact.
- **AI-driven prioritisation** — risk scoring lets Geldium prioritise outreach, cut manual triage, and improve repayment behaviour.
- **Cost-effective and scalable** — SMS and email are cheap and easy to scale.
- **Strategically aligned** — consistent with the company's customer engagement goals.

3. Ethical and Responsible AI Considerations

3.1 Fairness Risks and Mitigation

Risk	Description	Mitigation
Demographic Bias	The model may unfairly flag certain age or income groups as high risk	Regularly evaluate model performance across demographic segments using fairness metrics such as the **disparate impact ratio**
Feature Bias	Over-reliance on correlated features (e.g., employment gaps) could reinforce systemic inequality	Use **interpretable models** and monitor feature importance with tools such as **SHAP values**

3.2 Model Transparency

To ensure responsible deployment:

- **Plain-language explanations** for stakeholders, for example:
  "*This customer is flagged due to high credit usage and a recent missed payment, which are linked to higher default risk.*"

- **Explainability by design** — use transparent models (e.g., logistic regression) and keep **audit trails** for every decision.

Conclusion

The predictive model identifies **actionable risk signals** that can guide collection strategy with greater precision. By focusing on high-risk segments and applying ethical AI practices, Geldium can **reduce delinquency while maintaining fairness and trust**. The proposed intervention is data-informed and aligned with business priorities.

Key takeaways

- Highest-risk segments: utilisation >90%, repeat missed payments, under-30s with thin credit files
- Strongest short-term predictor: missed payments in the last 3 months
- Action: 6-week SMS + email campaign → target 10% drop in 30-day delinquency for the segment
- Guardrails: fairness checks (disparate impact), SHAP monitoring, plain-language explanations, audit trails